Natural Language Inference

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What is Natural Language Inference?

Natural Language Inference (NLI) determines whether a hypothesis is supported or contradicted by a premise.

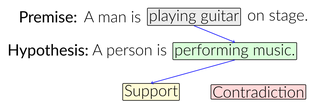

Figure 1. NLI determines the relationship between a premise and a hypothesis

Dataset Information:

- Binary classification task (support vs contradiction)
- 24,000 premise–hypothesis pairs for training
- 6,736 premise–hypothesis pairs for validation

Project Overview

This project explores two complementary approaches to the NLI task:

1. **ESIM-lite BiLSTM with Soft Attention:** A lightweight neural model using FastText embeddings, shared BiLSTM encoding, attention-based alignment, and comparison features.
2. **Transformer Ensemble:** A RoBERTa-based system combining a cross-encoder and bi-encoder through weighted ensembling to improve prediction performance.

These approaches enable a comparison between efficient sequence-based models and powerful transformer architectures for sentence-pair reasoning.

Achievements

Both approaches outperform their respective baselines:

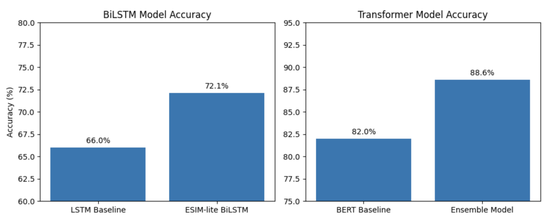

Figure 2. Performance comparison of Our Approaches vs Baseline

Key Findings & Error Analysis

- Attention and comparison mechanisms in the BiLSTM model help capture alignment and semantic relationships between sentences.
- The cross-encoder (Macro-F1 ≈ 0.88) outperforms the bi-encoder (≈ 0.79), showing the benefit of joint modelling.
- The ensemble further improves performance (Macro-F1 ≈ 0.883, $\alpha = 0.8$), indicating complementary strengths.
- Errors are balanced across classes (423 vs 360), indicating no strong prediction bias, with slightly higher recall for Class 1 (0.90 vs 0.87) suggest uneven class detection.
- Remaining errors (~12%) suggest difficulty with semantically subtle sentence pairs.

ESIM-lite BiLSTM with Soft Attention

Architecture Overview

Premise and hypothesis are encoded using FastText embeddings and a shared BiLSTM. Attention aligns relevant words, while comparison features capture difference and similarity. The representation is pooled and classified to produce the final prediction.

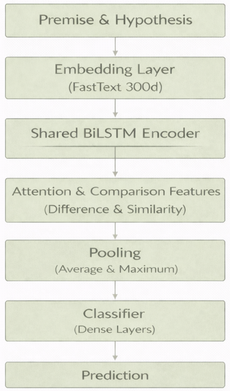

Figure 3. BiLSTM with attention and comparison for NLI

ESIM-inspired Components

Our approach incorporates key ideas from ESIM for sentence interaction:

- **Contextual Encoding:** Shared BiLSTM encodes premise and hypothesis
- **Soft Attention:** Aligns relevant words across sentences
- **Comparison Features:** Captures difference and similarity
- **Aggregation:** Uses average and max pooling

Example Prediction

- **Premise:** A man is playing guitar on stage.
- **Hypothesis:** A person is performing music.
- **Prediction:** Support

Aligned concepts: playing guitar ↔ performing music

RNN Model Results

Our BiLSTM with attention model improves over the provided baseline:

- **Baseline LSTM:** 66% accuracy
- **Proposed Model (BiLSTM + Attention):** 72.1% accuracy (+6.1% over baseline)
- **Proposed Model (BiLSTM + Attention) Macro F1:** 72.1%

The use of attention and comparison features improves the model's ability to capture semantic relationships between premise and hypothesis.

Transformer Model Overview

Architecture Overview

Our transformer-based system uses RoBERTa for the NLI task. Our approach combines two models:

- Cross-encoder for detailed interaction modelling
- Bi-encoder for semantic similarity

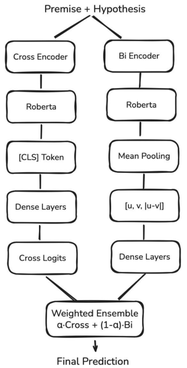

Figure 4. RoBERTa based transformer model for NLI Task C

These models are combined using a weighted ensemble to improve performance and robustness.

Transformer Model - RoBERTa Based Approach

Cross-Encoder

- Takes premise and hypothesis together
- Captures detailed word-level interactions
- Uses [CLS] representation for classification

Bi-Encoder

- Encodes sentences separately
- Uses mean pooling to obtain embeddings
- Combines:
  - u (premise)
  - v (hypothesis)
  - $|u - v|$

Ensemble

- Final = $\alpha \times$ Cross + $(1 - \alpha) \times$ Bi
- Best $\alpha = 0.8$

Transformer Model Results

- **Cross Encoder:** 88% accuracy. Strong performance on sentence interactions
- **Bi-Encoder:** 79% accuracy. Captures semantic similarity.
- **Ensemble:** 88.6% accuracy, 0.886 macro F1

Limitations and Ethical Considerations

- Input truncation may cause loss of context; limited training may affect optimisation.
- **Ethical:** Model predictions may reflect bias and should not be used in high-stakes settings without human oversight.

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